



JOHNS HOPKINS

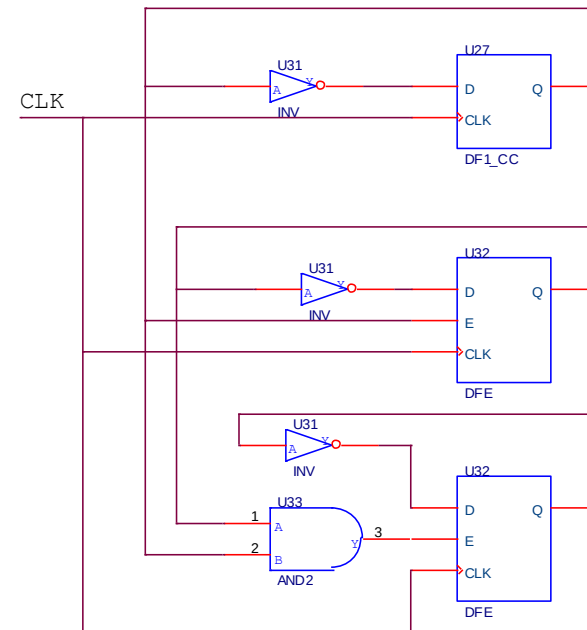
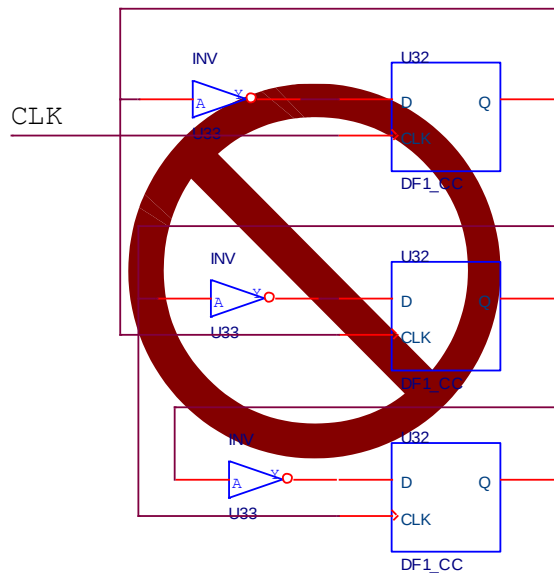
WHITING SCHOOL
of ENGINEERING

FPGA Design Using VHDL

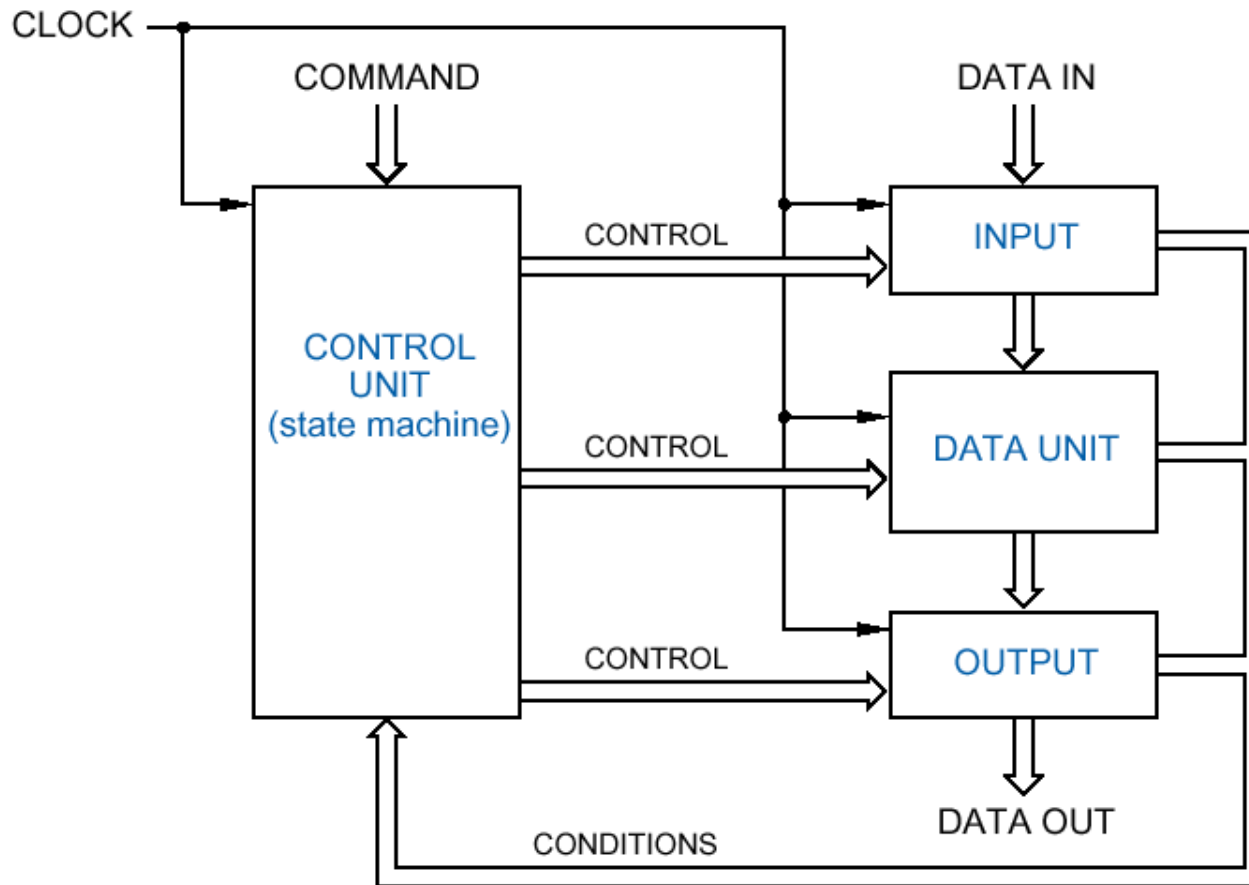
Module 3C: Synchronous Designs and Timing

Synchronous Design

- In a purely synchronous system, all flip-flops in the design are clocked by the same clock and asynchronous reset
 - Asynchronous resets are used for initialization

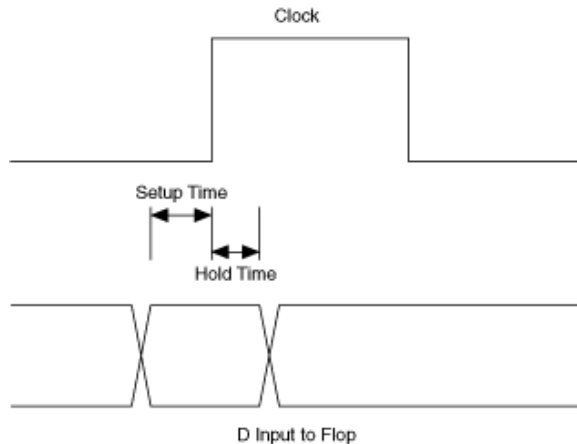


Typical Synchronous System



DFF Realm: Setup/Hold Time

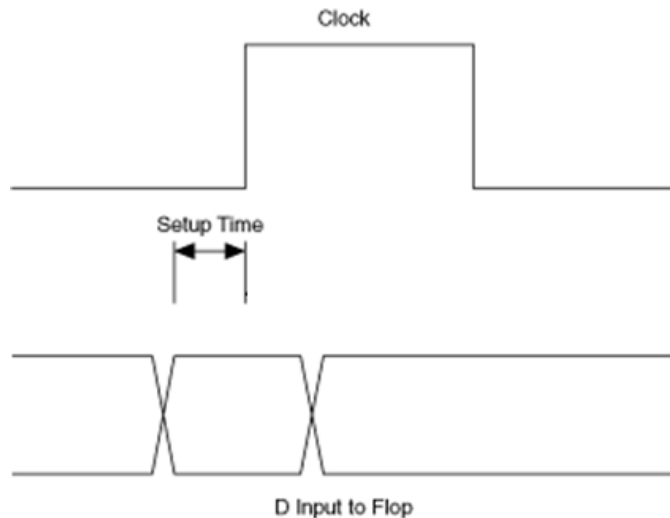
- “Setup Time” and “hold time” describe the timing **requirements on the data input** of the flip-flop or register with **respect to the clock**



Source: The Ultimate Hitchhiker's
Guide to Verification

Setup Time

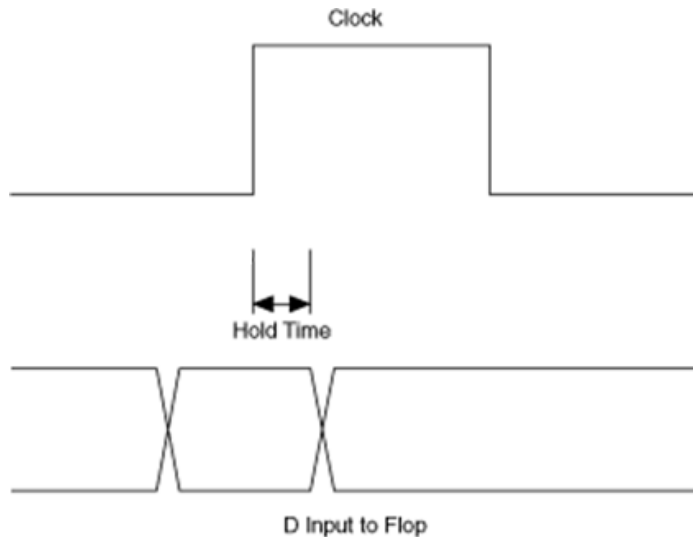
- A positive **setup** time describes the length of time that the **data** (to be clocked into the flip-flop) must be available and **stable before** the active **clock** edge



Source: The Ultimate Hitchhiker's
Guide to Verification

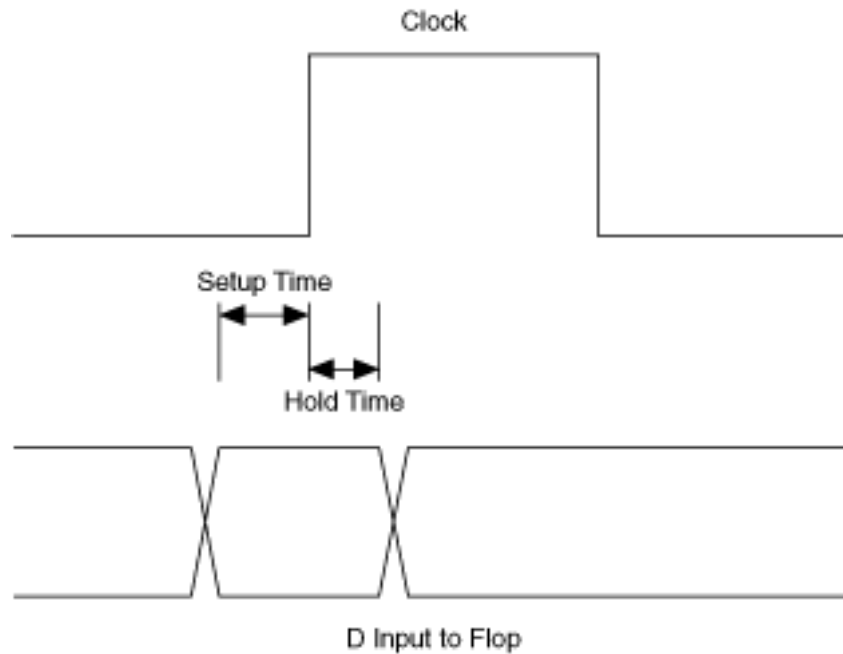
Hold Time

- A positive **hold** time describes the length of time that the **data** (to be clocked into the flip-flop) must remain available and **stable after** the active **clock** edge



Source: The Ultimate Hitchhiker's
Guide to Verification

Setup/Hold Time



Source: The Ultimate Hitchhiker's Guide to Verification

- Good quiz question

Clock Skew

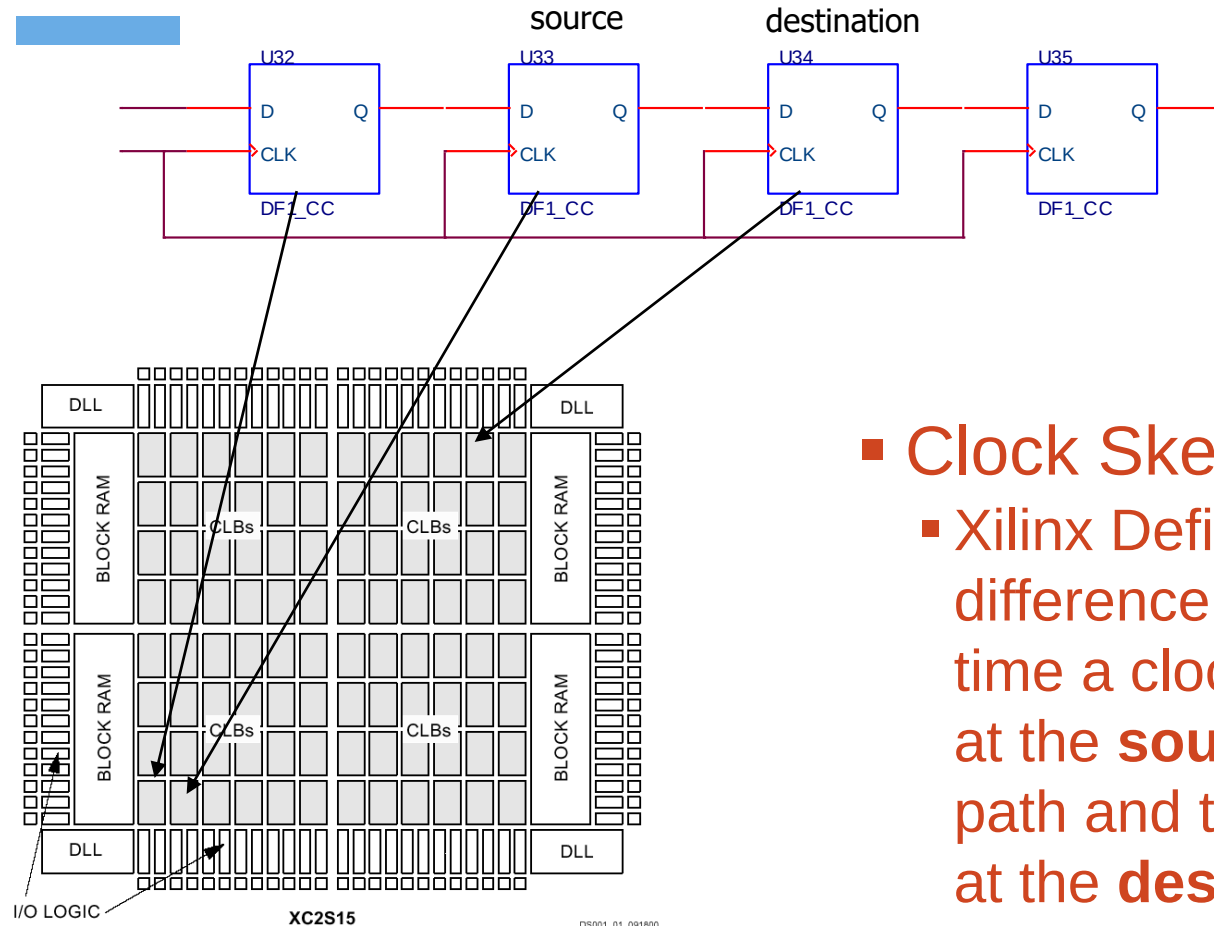


Figure 1: Basic Spartan-II Family FPGA Block Diagram

- Clock Skew
 - Xilinx Definition: The difference between the time a clock signal arrives at the **source** flip-flop in a path and the time it arrives at the **destination** flip-flop.

Figure Source: Spartan-II FPGA Family Data Sheet, DS001 June 13, 2008

Global Clock Buffers

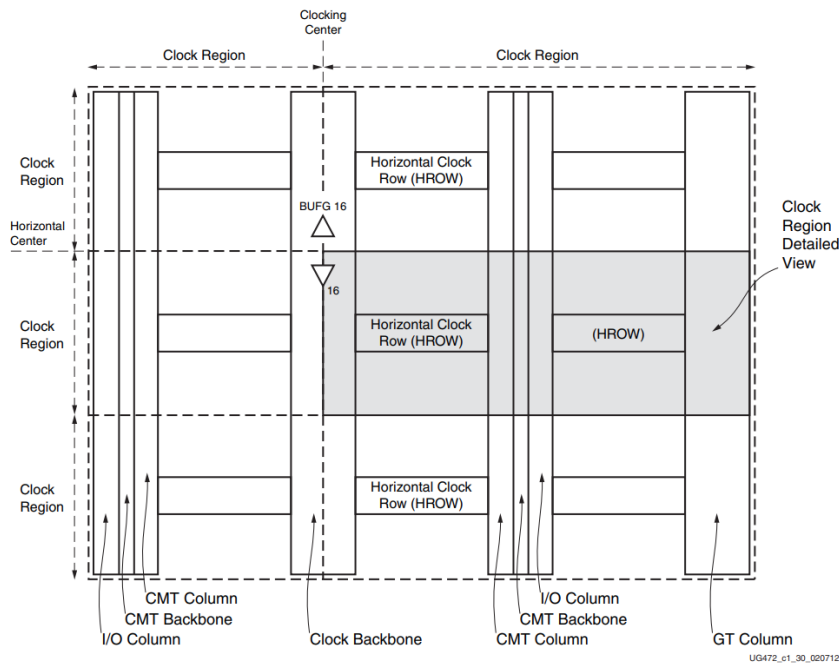


Figure 1-1: 7 Series FPGA High-Level Clock Architecture View

- Xilinx provides buffers to reduce clock-skew
- Signals on these buffers come into FPGA on special pins or are buffered from the logic via special cell

Site Type	Used	Fixed	Prohibited	Available	Util%
BUFGCTRL	1	0	0	32	3.13

Figure Source: 7 Series FPGAs Clocking Resources User Guide, UG472 (v1.14) July 30, 2018